06	AST-401	Operations Research OEC-1	**HSMC (OEC I)**	3	3	-	-	30	45	75
07	AST-402	Economics OEC-2	**HSMC (OEC II)**	3	3	-	-	30	45	75
PRACTICAL (LAB.)										
I	CSL-401	DBMS Lab - PCL5	**PCC**	1	-	-	2	15	10	25
II	CSL-402	Python Programming Lab – PCL6	**PCC**	1	-	-	2	15	10	25
III	CSL-403	Operating System Lab – PCL7	**PCC**	1	-	-	2	15	10	25
IV	ASL-401	Numeric and Scientific Computing Lab.	**ESC**	2	-	-	4	30	20	50
		Total		**22**	**19**	**-**	**10**	**245**	**305**	**550**

B. TECH. COMPUTER ENGINEERING
COURSE STRUCTURE
B. TECH. COMPUTER ENGINEERING -III YEAR (Effective from July 2025)

S.No.	Course No.	Course Name	Type of Course	Credit	Periods/week			Examination Scheme (Distribution of Marks)				
								Mid Semester Evaluation			End Semester Evaluation	Total Marks
					L	T	P	CCA	MTE-1	MTE-2		
THEORY			**Fifth Semester**									
01	CSE-501	Introduction to Data Mining – PEC1	**PEC**	3	3	-	-	30			45	75
02	CSC-501	Automata Theory – PCC8	**PCC**	3	3	-	-	30			45	75
03	CSC-502	Microprocessor – PCC9	**PCC**	3	3	-	-	30			45	75
04	CSC-503	Computer Networks - PCC10	**PCC**	3	3	-	-	30			45	75
05	CSC-504	Software Engg - PCC11	**PCC**	3	3	-	-	30			45	75
06	CSC-505	Object Oriented Programming – PCC12	**PCC**	3	3	-	-	30			45	75
PRACTICAL (LAB.)												
I	CSL-501	OOPs Lab - PCL8	**PCC**	1	-	-	2	15			10	25
II	CSL-502	Microprocessor Lab - PCL9	**PCC**	1	-	-	2	15			10	25
III	CSL-503	Computer Network Lab - PCL10	**PCC**	1	-	-	2	15			10	25
IV	CSL-504	Software Engg. Lab - PCL11	**PCC**	1	-	-	2	15			10	25
		Total		**22**	**18**	**-**	**8**	**240**			**310**	**550**

CSE- 501: INTRODUCTION TO DATA MINING

L	T	P	Internal: 30 Marks
2	1	0	External: 45 Marks
Credits : 3			Total: 75 Marks

Duration of Exam : 3 Hours

Course Outcomes:

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1. Understand the data mining concepts and the various tasks related to data mining.
2. Apply preprocessing methods in order to prepare data for applying various data mining task.
3. Analyze and evaluate various association rule mining algorithms and should write algorithms to implement it.
4. Analyze various classification algorithms and should be able to apply the algorithms on different datasets.
5. Apply various clustering algorithms on real and large datasets. Students should be able to take research level problems related to data mining and should be able to implement it.

UNIT 1:

Introduction to Data Mining: KDD, Process and Data Mining; KDD Steps; Types of Data for Data Mining, Data Mining Functionalities, Overview of data mining techniques.

Data Preprocessing: Introduction to Data Preprocessing; Descriptive Data Summarization: Measuring and Central Tendency and Dispersion of Data; Visualization of Descriptive Data Summaries; Data Cleaning: Handling Missing Values, Filtering Noisy Data – Binning Method; Data Integration; Data Transformation: Smoothing, Aggregation, Generalization, Normalization and Feature Selection; Data Reduction. Corerelation analysis

UNIT 2:

Association Rule Mining: Market basket Analysis; Frequent Item sets, Closed Item sets, and Association Rules; Support and Confidence; Apriori Algorithm for Mining Frequent Itemsets Using Candidate Generation; Generating Association Rules from Frequent Item sets; Improving the Efficiency of Apriori Algorithm; FP-Growth Algorithm for Mining Frequent Item sets without Candidate Generation; Mining Closed & Max Frequent Itemsets;

UNIT 3:

Basic Classification: Introduction to Classification and Prediction; Classification by Decision Induction; Attribute Selection Measures: Information Gain, Gain Ratio, and Gini Index; Tree Pruning; Bayesian Classification: Bayes'Theorem, Naïve Bayesian Classification

UNIT 4:

Advanced Classification: Bayesian Belief Networks; Classifier Accuracy Measures: Sensitivity, Specificity, Precision, and Accuracy; Predictor Error Measures; Accuracy Evaluation Methods: Holdout, Random Sub sampling, Cross-validation, and Bootstrap

Bagging and Boosting; Lazy Learners: K-Nearest- Neighbour Classifier; Prediction: Introduction to Linear and Non-Linear Regression.

UNIT5:

Cluster Analysis: Introduction to Cluster and Clustering; Features Required for Clustering Algorithms; Data Types and Dissimilarity Measures in Cluster Analysis; Categorization of Clustering Methods; Partitioning-Based Clustering: k-means Algorithms, k-Medoids algorithms, Hierarchical Clustering: Agglomerative and Divisive Methods.
Introduction to Web Mining and Text mining, Problem discussion.

Recommended Books:

- Data Mining by Hanand Kamber, Elsevier Publication.
- Introduction to Data Mining by Tan, Steinba chand Kumar, Pearson Publication.
- Practical Machine Learning Tools and Techniques with Java Implementations by H. Witten and E. Frank Morgan Kaufmann.
- Advances in Knowledge Discovery and Data Mining by U. M. Fayyad, G. Piatetsky-Shapiro, P. Smyth, and R. Uthurusamy, MITPress.

CSC- 501: AUTOMATA THEORY

L T P	**Internal: 30 Marks**
2 1 0	**External: 45 Marks**
Credits : 3	**Total: 75 Marks**
Duration of Exam : 3 Hours	

Course Outcomes:

1. Design different types of Finite Automata, their equivalence and applications.
2. Design Regular expressions from Finite Automata and solve problems using pumping lemma for Regular Languages.
3. Understand the Chomsky classification of Languages. Design Context Free Grammars for languages along with their simplification.
4. Design PDA for various problems, construction of PDA from a CFG along with the concept involving use of PDA to build Parsers.
5. Design Turing machines for various class of problems.

UNIT 1:

INTRODUCTION TO FINITE AUTOMATA: Introduction to Finite Automata, strings, alphabets and languages, graphs & trees, state tables & diagram, NDFA & DFA concepts, Conversion of NFA to DFA, Minimization of FA, Mealy & Moore machines.

UNIT 2:

REGULAR EXPRESSION: Introduction to Regular Expressions, Identities for Regular expressions, Arden's Theorem, Conversion of Finite Automata to Regular Expression, properties of regular languages. Pumping Lemma for Regular sets.

UNIT 3:

CONTEXT FREE GRAMMARS& LANGUAGES: Introduction to Grammars and Languages, Chomsky Classification of languages, Context free Grammar, Left Most & Right Derivations, Derivation tress, Ambiguity, Simplification of CFG, Conversion to Chomsky Normal Form.

UNIT 4:

PUSH DOWN AUTOMATON: Introduction to Push Down Automaton (PDA) for Context Free languages, Basic Design of a PDA, Instantaneous configuration of PDA, Construction of PDA for Context free languages, conversion from CFG to PDA.

UNIT 5:

TURING MACHINES &COMPUTABILITY: Introduction to Turing Machines (TM), Computing with Turing Machines, Non-deterministic TM, Primitive Recursive functions**,** Halting Problem of the TM, Computability and Types of Problems in Computer Science.

Books:

- J.E. Hopcroft & J.D. Ullmann, "Introduction to Automata Theory Language and Computation", Narosa Publications.
- K. L. P. Mishra & Chandrasekaran, “Theory of Computer Science: Automata, Languages and Computation”, 3rd Edition, PHI
- H.R. Lewis & C.H. Papadimitrou, "Elements of the Theory of Computation", PHI
- John C. Martin, "Introduction to Languages and the Theory of Computation", McGraw-Hill International
- D.A. Cohen, "Introduction to Computer Theory", John Wiley.

CSC- 502: MICROPROCESSOR

L	T	P	Internal: 30 Marks
2	1	0	External: 45 Marks
Credits : 3			Total: 75 Marks
Duration of Exam : 3 Hours			

Course Outcomes:

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1. Students will be able to understand Timing diagram and architecture of 8085, 8086 and other microprocessors.
2. Students will be able to write and use the instructions in Instruction set of 8085, 8086 correctly.
3. Students will be able to write 8085 programs.
4. Students will have understanding of Interfacing and Interrupt of 8085.
5. Students will be able to develop 8086 Programs, software interrupt and can interface peripheral devices with 8086.

UNIT 1:

Review, Organization and architecture of 8085,8086 and other Microprocessors, Instructions of 8085 & Programming techniques, Machine Language Vs Assembly Language, Basic concepts of timing & control unit, Timing Diagrams for 8085 instructions. Study of important 8 – bit Microprocessors & their Comparison, Introduction to 16 – bit processors – 8086, 8088 and 68000 Coprocessor & comparison. Introduction to 32 – bit Microprocessors.

UNIT 2:

Instruction set of 8085, instruction set of 8086, Syntax and illustration of different types of instructions along with their use.

UNIT 3:

Programs using instruction set of 8085, maximum, minimum, searching ,sorting, division, multiplication, factorial ,conversion from binary to decimal and ASCII, decimal to Binary , matrix addition and other relevant programs using instruction set of 8085, Subroutine, Counter, delay calculations,

UNIT 4:

Minimal System, Necessity for interfacing, Address space partitioning – Memory mapped I / O & I / O Mapped I / O, Advantages and Disadvantages, Types of Interfacing devices – I / O ports, Programmable peripheral interfaces 8255, Various interrupt Schemes, Multiple Interrupt, Enabling, Disabling and Masking of Interrupts Particularly in 8085, DMA & DMA Controller.

UNIT 5:

TASM, MASM, programs based on 8086 instruction set, Software Interrupt, Assembly language programming in C, Interfacing with 8086, Microprocessors based system design

Books:

- K.M. Bhurchandi and A.K. Ray, Advanced Microprocessor and Peripherals, Tata McGraw Hill
- A.P. Mathur, "An Introduction to Microprocessors" Tata McGraw Hill, 1995.
- K.L. Short, "Microprocessor & Programmed Logic", 2nd Ed., PHI, 1994
- R.G. Gaonkar, "Microprocessor Architecture programming and application", Wiley Eastern Ltd., 1994.

CSC- 503: COMPUTER NETWORKS

L	T	P	Internal: 30 Marks
2	1	0	External: 45 Marks
Credits : 3			Total: 75 Marks
Duration of Exam : 3 Hours			

Course Outcomes:

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1. Students will be able to memorize and remember the Fundamentals of Computer Networks.
2. Student will be able to understand various switching techniques, Communication media, and Digital Analog formats.
3. Student will be able to Apply Flow Control Algorithms and Evaluate the Error Detection and Correction Methods.
4. Student will be able to Analyze the performance of the Computer Networks.

5. Student will be able to configure and create Routing Tables and Private Networks.

UNIT 1:

Introduction:

Introduction to Computer Networks, LAN, MAN, WAN, Uses of Computer Networks, LAN Technologies- Transmission Topologies, Access methods. Network Architecture, Protocol and standards, References Model: OSI-ISO and TCP/IP, Overview: Circuit switching, Message switching and Packet switching.

UNIT- 2

The Data Link Layer:

Data Link Layer design issues, Framing, Error Detection and Correction, Flow control Protocols, Stop and Wait protocol, Sliding - window flow control, Error control, stop and wait ARQ, Go-back-N, Selective repeat ARQ, Examples of Data link Protocols- HDLC.

UNIT- 3

The Medium Access Control Sub Layer:

The channel allocation problem, ALOHA, Multiple access Protocols, Collision free Protocols, IEEE Standards for LANs and MANs, Wireless LAN: IEEE 802.11, High speed LANs.

UNIT – 4

The Network Layer:

Addressing in IPV4: Classful addresses, CIDR notation, Classless addresses, Special addresses, Network Design: Sub-netting and Super-netting, Network Address Translation, IPV6.

UNIT – 5

Routing Algorithms-Dijsktra's, Distance vector: RIP, Link state: OSPF, BGP,Multicast Routing, and Hierarchical Routing. Delivering and forwarding of IP Packets, Datagram.

Books :

- B.A. Forouzan, " Data Communication and Networking", TMH, 5TH Edition.
- A.S. Tanebaum, " Computer Networks", 4th Edition Pearson Education.
- W. Stallings, " Data and Computer Communication", 7th Edition , Pearson Education.
- Comer E. Doughlas, " Computer Networks and Internet", 2nd Edition Pearson Education.

CSC- 504: SOFTWARE ENGINEERING

L	T	P	Internal: 30 Marks
2	1	0	External: 45 Marks
Credits : 3			Total: 75 Marks
Duration of Exam : 3 Hours			

Course Outcomes:

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1. Students will be able to understand fundamental theories of Software Development life cycle & various SDLC Models.

2. Student will be able to understand various types of Requirements, Techniques for Requirement Elicitation and SRS Document, COCOMO, Risk Management activities etc.

3. Student will be able to understand coupling and Cohesion and its types, Software Reliability and Quality assurance using ISO9001:2008, ISO9001: 2015 Standards and SEI-CMM.

4. Student will be able to understand various types of Software Testing and Generating test cases using various testing approaches and Debugging.

5. Student will be able to understand Software Maintenance, Types of maintenance, various models of maintenance, Software RE-ENGINEERING and REVERSE Engineering and Configuration Management & CASE Tool etc

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Unit 1: Introduction Definition, Program Vs Software, Software processes, Software life cycle models: Build and Fix, Waterfall, Prototype, Iterative Enhancement Model, Evolutionary and Spiral model, V Model & RAD Model.

Unit 2: Software Project Planning. Crucial process steps of Requirement Engineering, Types of requirements, Requirement Elicitation techniques and Requirement Documentation, SRS, COCOMO model, Risk management.

Unit 3: Software Requirement Analysis and Specifications, Design & Software Reliability. Problem Analysis, Data Flow Diagrams, use case diagrams, Software Prototyping, Cohesion & Coupling, Classification of Cohesiveness & Coupling, Function Oriented Design, Object Oriented Design, User Interface Design. Software Reliability: Failure and Faults, Overview of Quality Standards like ISO 9001, SEI-CMM

Unit 4: Software Testing Software Testing terminology, Functional testing: Boundary value analysis, Equivalence class testing, Cause- effect graphing, Structural testing: path testing, Data flow and mutation testing, unit testing, integration and system testing, Validation testing Debugging techniques & Approaches and Testing Tools.

Unit 5: Software Maintenance Software Maintenance & its types: Management of maintenance, The Maintenance Process, Maintenance Models: Quick fix, Iterative Enhancement, Reuse Oriented. Reverse Engineering, Software RE-engineering, Configuration Management.

BOOKS:

- Prof: KK Aggarwal & Yogesh Singh: SOFTWARE ENGG:
- Pankaj Jalote, " An Integrated Approach to Software Engg" Narosa Publishing House, New Delhi.
- Pressman,"Priciples of Software Engg" TMC, 5th Ed. 2005

CSC- 505: OBJECT ORIENTED PROGRAMMING

L	T	P	Internal: 30 Marks
2	1	0	External: 45 Marks
Credits : 3			Total: 75 Marks
Duration of Exam : 3 Hours			

Course Outcomes:

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1. Students will **define** and **critique** the fundamental features of object oriented programming.

2. Students will **write** C++ programs to **demonstrate** polymorphism
3. Students will **explore** C++ programs with and without inheritance
4. Students will **develop** object-oriented programs in a new programming language
5. Students will **use** templated/generic components to **build** reusable software components

UNIT 1: OBJECT ORIENTED PROGRAMMING USING C++

Object Oriented Paradigm, Structured vs Object Oriented Development, Concept of Object and classes, Encapsulation, Polymorphism, Inheritance, Merits and demerits of OOP. Class specification, Class objects, Defining member function, Inline functions, Data Hiding, Empty class, Passing objects as parameters, Returning objects from functions, Static data and member functions. Constructors and destructors, overloading of constructors, Dynamic initialization through constructors, Copy constructors, Static data members with constructors and destructors. Pointers to objects, Array of objects, this pointer

UNIT 2: IMPLEMENTING POLYMORPHISM IN C ++

Function and Operator overloading, Overloading of unary and Binary operators, Limitations of overloading of increment and decrement operators, overloading of arithmetic, Relational, assignment, new and delete, subscript operators. Data conversion between objects. Complete conversion. Overloading through friend functions. Tracing of memory leaks.

UNIT 3: INHERITANCE AND VIRTUAL FUNCTIONS

Declaration of derived class, forms of inheritance, constructors and destructors in derived class, types of inheritance, abstract class, Virtual functions: Need of virtual functions, Pointers to derived class objects, pure virtual functions, Virtual destructors, Rules of writing virtual function

UNIT 4: OBJECT ORIENTED PROGRAMMING USING JAVA

Classes, objects and constructor in Java, Implementing inheritance and polymorphism - dynamic binding, method overriding, abstract classes and methods. Interfaces - Interfaces vs. Abstract classes, defining an interface, implementing interfaces, accessing implementations through interface references, extending interfaces.

UNIT 5: GENERIC PROGRAMMING IN JAVA

Introduction to Exception Handling, Generic programming in Java. Why Use Generics? Types of Java Generics. Generic classes. Generic functions. Generics, Inheritance, and Subtypes. Type inference. Wildcards. Type erasure.

BOOKS

1. The C++ Programming Language by B.Stroustrup, Pearson Education.
2. Thinking in C++ by Bruce Eckel, Pearson Education
3. Object Oriented Programming in C++ by N.Barkakati, PHI
4. Mastering C++ by Venugopal and et all, Tata McGraw Hill
5. C++ How to Program by Deitel and Deitel, Pearson Education
6. The Complete Reference Java by Herbert Shildt, Tata McGraw Hill
7. Object-Oriented Programming in C++ by Robert Lafore, 4th Edition, Pearson Education.
8. Java Generics and Collections: Speed Up the Java Development Process, 1st Edition. O'Reilly.

CSC- 601: ANALYSIS AND DESIGN OF ALGORITHMS

L	T	P	
2	**1**	**0**	**Internal: 30 Marks**
			External: 45 Marks